

Competition and coexistence

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Learning objectives

1. Know the processes that lead to species coexistence
2. Add a competitor to the Beverton-Holt model
3. Perform an invasion analysis

Competition and coexistence

Interspecific competition and its outcomes

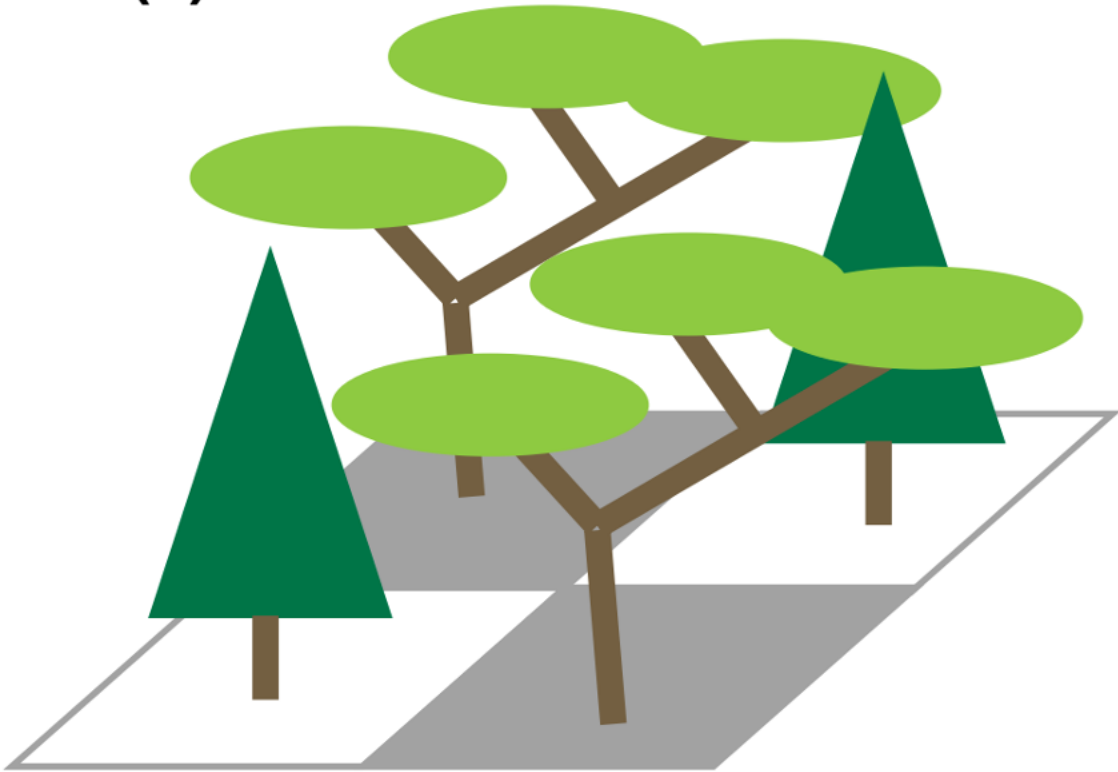
- interspecific competition: one species has reduced growth rate in presence of another species
- outcomes:
 - competitive exclusion - one species outcompetes other species
 - coexistence - stabilize and both persist in long term *may have lower population growth or sizes compared to those without competitor, but if they persist, it's coexistence*
 - unstable coexistence - depends on species arrival *can coexist under the right initial conditions*

given that species compete with each other, how do you get diverse communities where many species coexist?

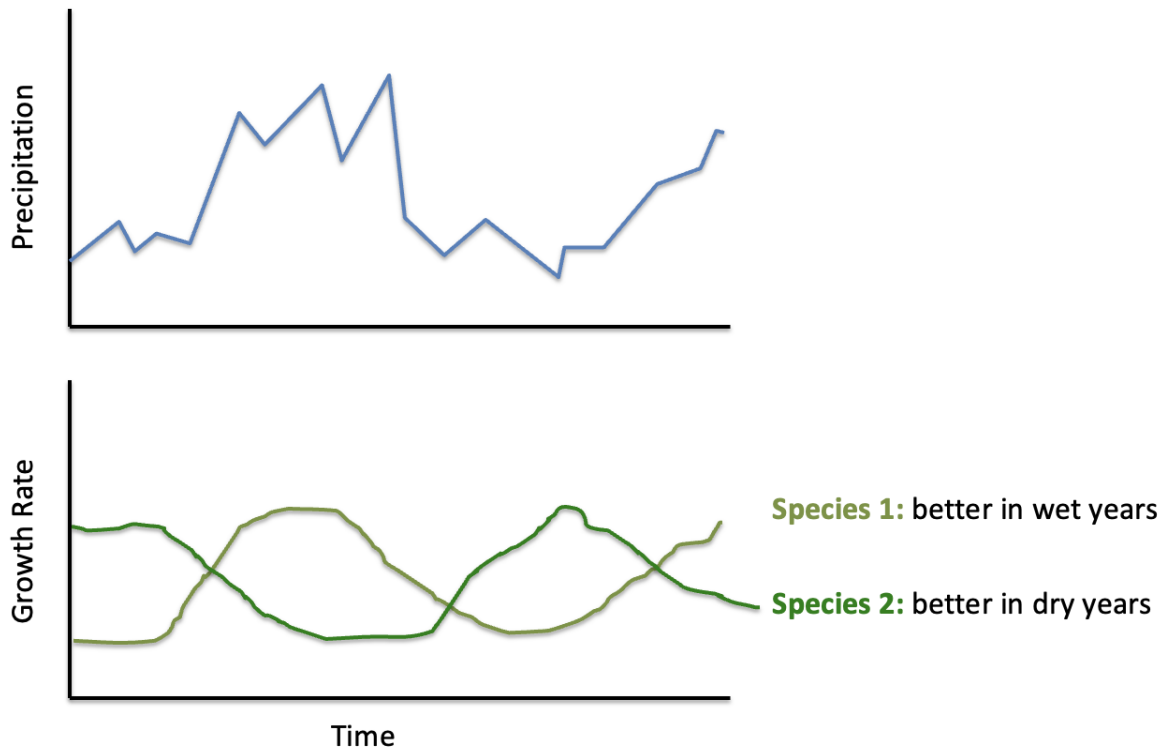
Processes leading to stable coexistence

- spatial heterogeneity - good and bad patches

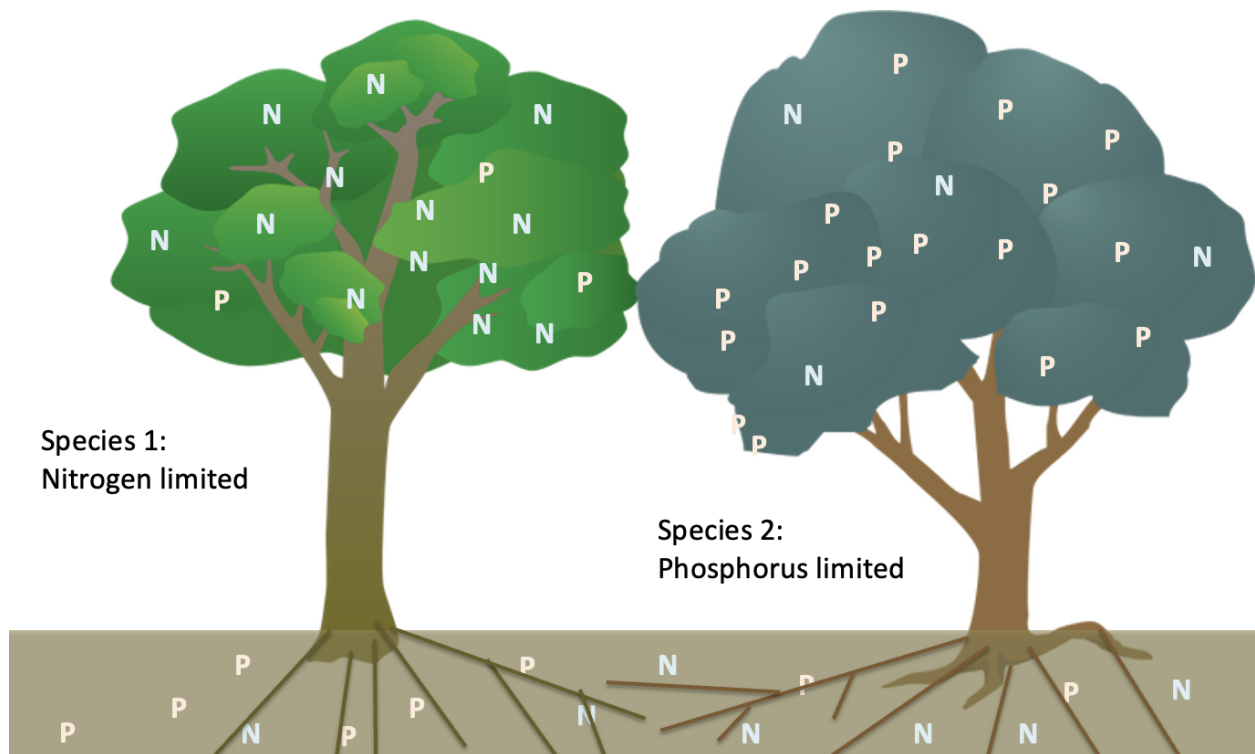
(a)



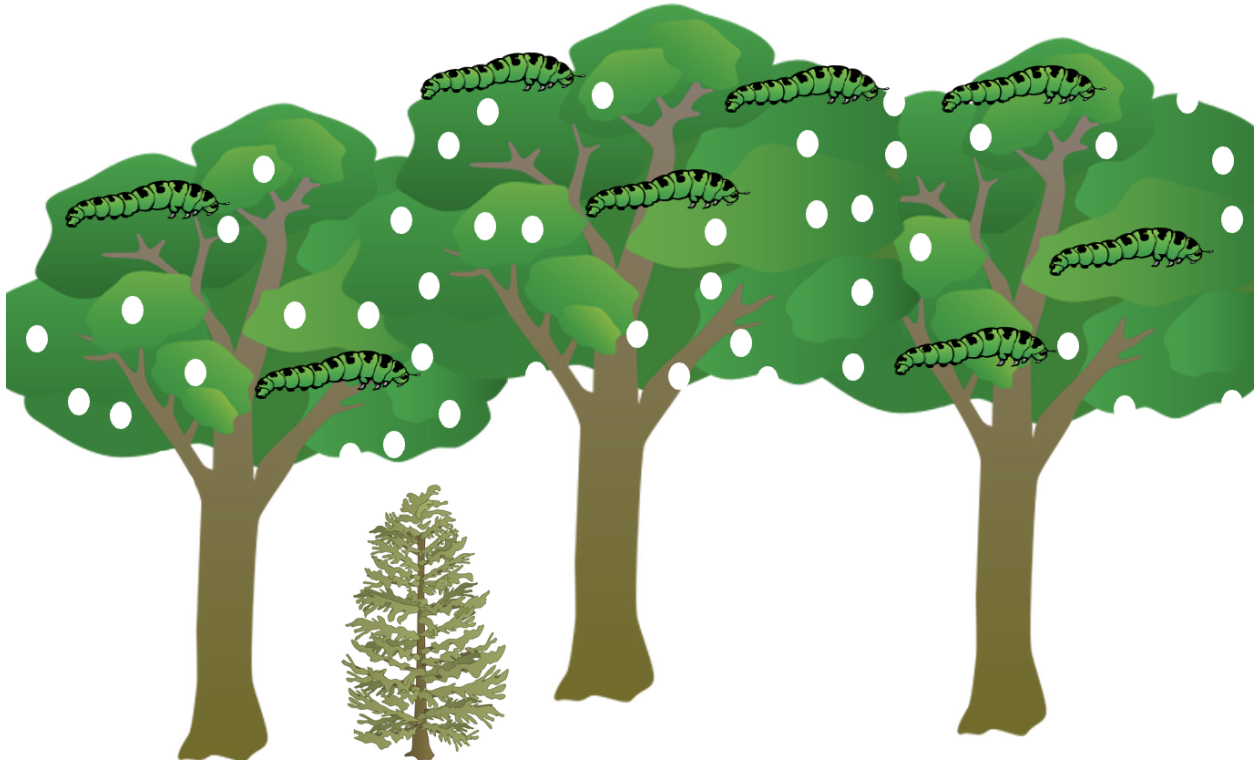
- temporal heterogeneity - good years and bad years



- resource partitioning - limited by different resources



- natural enemies - host-specific *talk about this in week 8*



species is limiting itself more than others- this makes sense because if multiple species are going to coexist, we can't have a species that has unrestricted growth, it'll just use up all the resources

- mechanisms lead to intraspecific competition > interspecific competition

Modeling interspecific competition

add to beverton-holt, previously:

$$N_{i,t+1} = \frac{\lambda N_{i,t}}{1 + \alpha_{ii} N_{i,t}}$$

add a second species to restrict population growth

- species i and j
- α_{ij} = effect of j on i

$$N_{i,t+1} = \frac{\lambda N_{i,t}}{1 + \alpha_{ii} N_{i,t} + \alpha_{ij} N_{j,t}}$$

we just added another term in the denominator, added subscripts to clarify which population we're modeling

Invasion analyses

how can we tell if species are coexisting?

plot1: one species goes to 0, one stabilizes

plot2: flat lines in parallel

plot3: stochastic lines in parallel

plot4: alternating species with greater abundance

this looks like coexistence, but what if transient dynamics, decline in future?

plot5: invasion, increase from low density

- both species increase from low density
- no environmental variation, need 1 time step
- environmental variation, need to average over conditions/years
- $\log \frac{N_{t+1}}{N_t} > 0$ for each species - coexistence, < 0 competitive exclusion